CHAPTER

24

Linear Image Processing

Linear image processing is based on the same two techniques as conventional DSP: *convolution* and *Fourier analysis*. Convolution is the more important of these two, since images have their information encoded in the spatial domain rather than the frequency domain. Linear filtering can improve images in many ways: sharpening the edges of objects, reducing random noise, correcting for unequal illumination, deconvolution to correct for blur and motion, etc. These procedures are carried out by convolving the original image with an appropriate filter kernel, producing the filtered image. A serious problem with image convolution is the enormous number of calculations that need to be performed, often resulting in unacceptably long execution times. This chapter presents strategies for designing filter kernels for various image processing tasks. Two important techniques for reducing the execution time are also described: *convolution by separability* and *FFT convolution*.

Convolution

Image convolution works in the same way as one-dimensional convolution. For instance, images can be viewed as a summation of *impulses*, i.e., scaled and shifted delta functions. Likewise, *linear systems* are characterized by how they respond to impulses; that is, by their *impulse responses*. As you should expect, the output image from a system is equal to the input image *convolved* with the system's impulse response.

The two-dimensional delta function is an image composed of all zeros, except for a single pixel at: *row* = 0, *column* = 0, which has a value of *one*. For now, assume that the row and column indexes can have both positive and negative values, such that the *one* is centered in a vast sea of zeros. When the delta function is passed through a linear system, the single nonzero point will be changed into some other two-dimensional pattern. Since the only thing that can happen to a point is that it *spreads out*, the impulse response is often called the **point spread function (PSF)** in image processing jargon.